Balancing Playback Continuity and Replica Recovery

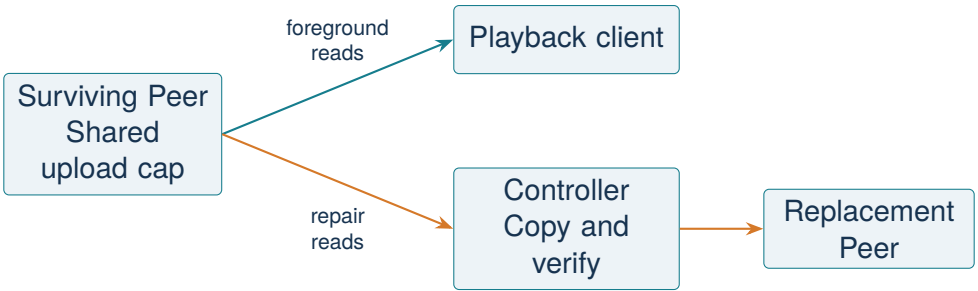
Cooperative Video Storage: System and Experimental Findings

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COMPSCI 512

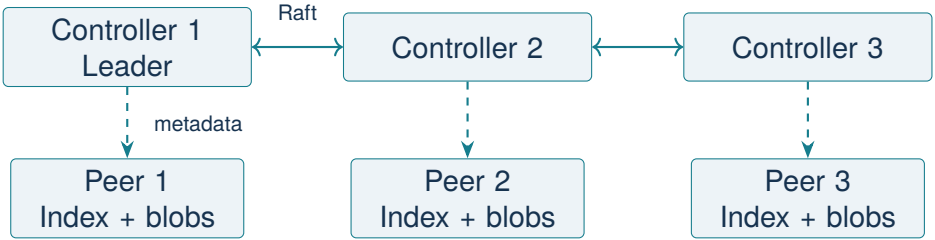
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One upload budget serves two demands



Repair restores a second source, but consumes bandwidth needed by viewers.

Control plane and data plane



- Raft: manifests, committed receipts, publication state, persistent tasks.
- Browser: Peer $\rightarrow$ Gateway $\rightarrow$ Browser.
- Experiment: Peer $\rightarrow$ headless client; Gateway is outside Raft.

Publish only after two durable copies

1. Create a **DRAFT** manifest.
2. Upload segments; Peers validate size and SHA-256 and persist bytes.
3. Confirm receipts from **two distinct holders per segment**.
4. Commit **READY**; expose the video for playback.

Raft commits metadata. Peers store the video bytes outside the Raft log.

Playback recovery $\neq$ redundancy recovery

Foreground recovery

Peer fails $\rightarrow$ try another holder $\rightarrow$ verify bytes $\rightarrow$ continue delivery.

Background repair

Detect shortage $\rightarrow$ grace interval $\rightarrow$ commit task
$\rightarrow$ copy and verify $\rightarrow$ commit replacement registration.

Continuing playback does not prove that redundancy has been restored.

Application implementation and validation

Application work	Foundation / tool
Publication and persistent task lifecycle	HashiCorp Raft, BoltDB, snapshots
Durable indexes and immutable objects	Go HTTP / filesystem, SHA-256
Playback, budgets and experiment harness	FFmpeg / FFprobe for media gates
Console: faults, cleanup and video deletion	React and TypeScript

Archived gates: build, static/race, smoke, repair, cleanup, deletion, exact fragment timing and media decoding.

Evidence: results/compact-study/validation/gates.json.

Optional demonstration: from upload to playback

1. Start three Controllers and three Peers.
2. Upload a rehearsed short clip; observe packaging and publication.
3. Play the video; inspect per-Peer segment counts.
4. Return to the controlled experimental results.

A functional demonstration and comparative performance evidence are different.

Three questions organize the study

RQ-A: interference and timing — E1 / E2

When does repair interfere with delivery? What does waiting change?

RQ-B: playback / recovery compromise — E3 / E4

Does load-aware repair improve outcomes over fixed controls?

RQ-C: correctness and recovery — E4 / E5

Which application properties survive Peer and Controller failures?

Every policy obeys the same total source cap

Policy	Treatment
N	No failure: normal-service reference
OFF	Failure without repair
S	Repair shares the common upload service
F25 / F50	Additional repair caps at 25% / 50% of source rate
L	Load-aware additional repair allowance

An additional cap limits repair; it does not reserve bandwidth or raise capacity.

L reacts to source-local foreground activity

Foreground busy

Repair allowance: **25%** of source cap.

Idle for 2 seconds

Relax allowance to **100%**.

Task age > 60 s

Repair allowance floor: at least **50%**.

Updates every 250 ms. Task age is persisted with the committed task.

The 60-second branch was regression-tested, but did not activate in these runs.

Independent processes on one shared host

Controlled setting	Executed values
Topology	3 Controllers + 3 Peers; real HTTP/TCP
Fixture	9 distinct 128 KiB segments; 2 logical seconds each
Source limits	96 / 144 / 192 KiB/s
Demand	Low: 2 viewers; high: 6; burst arrivals at 0 / 4 / 12 s
Client	Window 3; concurrency 2; startup buffer 4 s
Timeouts	8 s per download attempt; 42 s per session

Real byte transfer with simulated media consumption, not browser measurement.

Measure delivery and redundancy together

Executed study

81 unique runs in 67.9 minutes
72 E1–E4 runs + 9 E5 runs
396 scheduled viewer outcomes
Pilots excluded from estimates

Outcomes

Completion and observed stalls
Verified two-copy restoration
Repair payload
One-copy exposure (MiB·s)

First fault: 5 s. Observation horizon: 50 s. Recovery target: 40 s after fault. Three paired contexts; two per E4 cell.

Separate Peer loss from repair interference

N

No failure.
Normal-service reference.

OFF

Peer fails.
Repair disabled.

S

Same Peer failure.
Shared repair.

Compare low and high demand across three paired source-failure contexts.

OFF minus N exposes the fault; S minus OFF isolates the repair treatment.

Repair can add stalls under contention

Demand	Policy	Completed	Mean observed stall/viewer
Low	N / OFF / S	6/6 each	0.00 s each
High	N	18/18	3.20 s
High	OFF	7/18	3.24 s
High	S	7/18	4.46 s

Slowest-Peer failure: all six viewers finished; repair added **3.46 s/viewer**.

No added stalls at low demand; context-dependent interference at high demand.

Grace trades early action against unnecessary work

- Shared repair policy S; grace of **1 / 3 / 8 seconds**.
- Brief absence: original Peer returns after **4 seconds**.
- Persistent absence: original Peer stays offline.
- Measure repair payload and time spent with one reachable copy.

Grace controls when repair starts, not whether a started task is cancelled.

Waiting avoids copies, but extends exposure

Grace	Brief: repair MiB	Persistent: repair MiB	Persistent exposure MiB·s
1 s	0.125	0.750	17.42
3 s	0	0.750	18.24
8 s	0	0.750	20.20

All 54 brief-outage viewer sessions completed.

Persistent exposure increased about 16% from 1 s to 8 s grace.

Compare adaptation with credible fixed controls

Treatments

S: shared F25: 25% F50: 50% L: adaptive

- Steady high demand and burst arrivals.
- Same content, initial copies, source limits and paired fault trace.
- Evaluate completion, stalls and verified restoration together.

Less observed stall alone does not establish a better policy.

No consistent overall advantage for L

Policy	Steady stall	Burst stall	Verified by 40 s: steady / burst
S	4.46 s	6.18 s	3/3 2/3
F25	4.73 s	6.28 s	3/3 1/3
F50	4.88 s	6.20 s	3/3 2/3
L	4.33 s	6.21 s	3/3 1/3

All policies: **7/18** steady and **9/18** burst completions.

F25: one burst run unrecovered at the observation horizon.

The data do not establish a consistent L advantage over F50.

A local improvement has a recovery cost

Feasible high-load context: the slowest Peer fails; all six viewers finish.

Metric	F50	L
Observed stall per viewer	8.53 s	6.88 s
Verified restoration after fault	33.99 s	36.06 s
One-copy exposure	18.82 MiB·s	20.38 MiB·s

L saved 1.65 s of stalls, but restored redundancy 2.07 s later.

Can the replacement serve after both holders stop?

1. Stop the first original holder.
2. Wait a fixed **4 / 14 / 30 seconds**, independent of repair success.
3. Stop the second original holder.
4. Probe a preselected late-queue segment from its replacement.

S / F50 / L; two paired contexts per delay. Success requires correct hash *and* committed registration.

A buffered viewer may finish even when fresh delivery is unavailable.

The selected replacement was not ready in time

Policy	Second fault +4 s	+14 s	+30 s
S	0/2	0/2	0/2
F50	0/2	0/2	0/2
L	0/2	0/2	0/2

All 18 probes were unavailable under the specified success condition.

A boundary was observed; no policy advantage or permanent data loss was established.

Recovery requires a successful application request

Case	Passed	Median	Origin
E5a: active delivery	3/3	1.52 s	Leader stop
E5b: pending repair	3/3	1.60 s	Leader stop
E5c: majority loss	3/3	1.04 s	Quorum restored

Fresh quorum-backed metadata, not a health response or Leader role string.

All nine tested Controller scenarios passed the application contract.

E5a: fresh metadata recovered during delivery

- Stop the current Leader during active delivery.
- Probe fresh, quorum-backed metadata.
- Warm and cold uncached sessions completed in all three trials.

Observed recovery after Leader stop

Median **1.52 s**; observed range **1.47–2.74 s**.

E5b: committed repair survived Leader loss

- Stop the Leader with a committed pending repair task.
- New Leader retains the **same task ID and creation time**.
- Target bytes pass hash validation; replacement registration commits.
- Controller restart retains task state; source restart retains L configuration.

3/3 passed; fresh recovery median 1.60 s, range 1.52–1.97 s.

E5c: without a majority, writes were refused

Stable majority loss

9/9 valid write attempts returned HTTP 503.
No success acknowledgments.

After quorum restoration

Baseline state retained.
Original operation IDs reconciled.
Median recovery: 1.04 s.

Recovery range 0.85–1.43 s, measured from quorum restoration.

Three findings, with explicit boundaries

RQ-A

Repair can compete with playback. Grace avoids some brief-outage work but extends reduced-redundancy exposure.

RQ-B

L showed a local compromise, not a consistent advantage over F50.

RQ-C

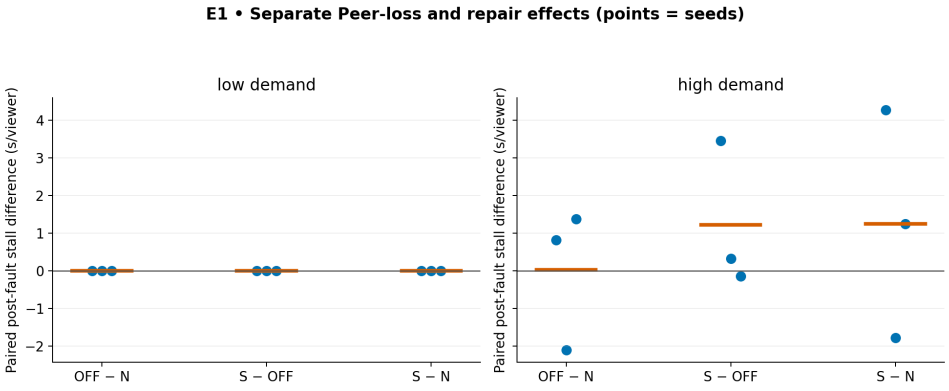
Application recovery and task continuity held in the tested Controller failures. E4 exposed unfinished-repair availability limits.

Limits and the next experiments

Current boundary	Next experiment
One host; shared hardware and network stack	Separate hosts; controlled delay/loss
Three contexts; two per E4 cell	More repetitions and capacity strata
Headless consumption; short horizon	Browser measurements; long-lived task aging
E4 target never ready at tested delays	Later probes; verified-completion control

We characterize a tradeoff; we do not claim universal adaptive-policy superiority.

E1: paired effects



Exploratory • paired seeds 1101,1102,1103 • 3 Controllers / 3 Peers • real HTTP, simulated consumption
50 s nominal horizon; 2 s verification cadence; payload counters include retries; see raw journals and provenance.

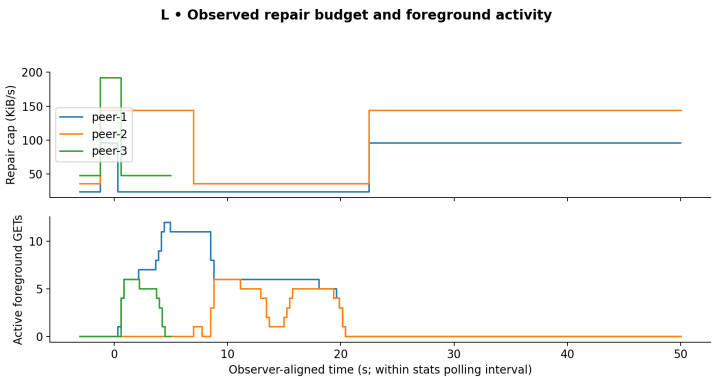
E3: outcomes by failed source capacity

E3 • high demand: retain the failed-source context

Failed Peer	Policy	Viewers done	Observed stall (s / scheduled viewer)	Restoration interval (s after fault)	Verified by 40 s
peer-1	S	6/6	7.47	(33.95, 35.99]	yes
peer-1	F25	6/6	8.36	(35.77, 37.77]	yes
peer-1	F50	6/6	8.53	(31.89, 33.99]	yes
peer-1	L	6/6	6.88	(33.99, 36.06]	yes
peer-2	S	1/6	1.83	(32.02, 34.03]	yes
peer-2	F25	1/6	1.61	(37.86, 39.91]	yes
peer-2	F50	1/6	1.77	(31.77, 33.77]	yes
peer-2	L	1/6	1.77	(31.75, 33.79]	yes
peer-3	S	0/6	4.07	(29.76, 31.79]	yes
peer-3	F25	0/6	4.21	(38.00, 39.99]	yes
peer-3	F50	0/6	4.33	(31.92, 33.93]	yes
peer-3	L	0/6	4.33	(32.07, 34.13]	yes

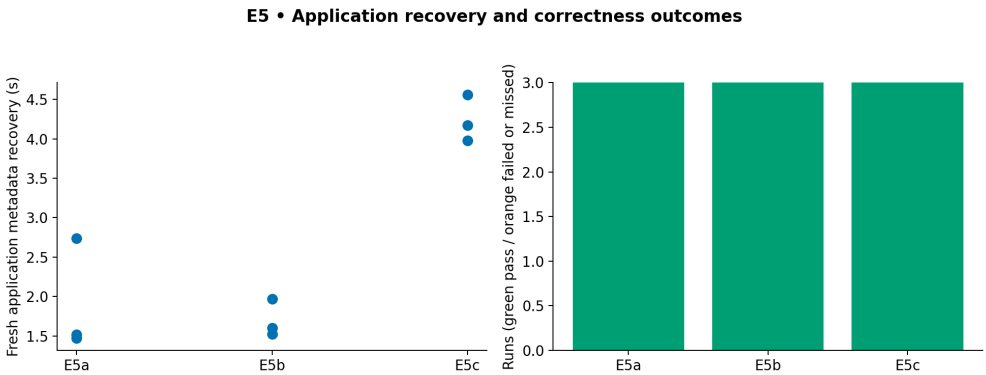
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50 s nominal horizon; 2 s verification cadence; payload counters include retries; see raw journals and provenance.

L: recorded repair-budget behavior



Trace: peer-high-L-3-persistent-0-1103 • 0/6 viewers completed; foreground idle here follows viewer failures.
Lines show recorded 250 ms source control updates, aligned within the stats interval; all classes still obey the common source upload cap.

E5: fault-relative recovery distribution



Exploratory • 3 trials per case • direct application probes • E5c latency includes deliberate quorum isolation; see E5-safety-table for restoration-relative timing.